

Winter Crossword Solutions

Aaron Qin, Sohun Uttamchandani, Kevin Zhu

1 1	2 3
3 6	8

ONE ACROSS

Complementary counting with casework works well here.

In total, there are

$$\binom{4}{2} = 6$$

different roads.

If all 6 roads are blocked, there is clearly no way to travel between the towns. The probability of this happening is

$$\left(\frac{1}{2}\right)^6 = \frac{1}{64}.$$

If all but 1 or 2 road(s) are blocked, there is also no way to travel between the towns. The probability of this happening is

$$\binom{6}{1} \left(\frac{1}{2}\right)^5 \left(1 - \frac{1}{2}\right) + \binom{6}{2} \left(\frac{1}{2}\right)^4 \left(1 - \frac{1}{2}\right)^2 = \frac{21}{64}.$$

The only combinations of 3 blocked roads that result in Santa not having a connected path occur when all three of these roads are incident to a common town. The probability of this happening is

$$\binom{4}{3} \left(\frac{1}{2}\right)^3 \left(1 - \frac{1}{2}\right)^3 = \frac{4}{64}.$$

Adding these probabilities, we find that Santa has a

$$\frac{1 + 21 + 4}{64} = \frac{26}{64} = \frac{13}{32}$$

chance of not being able to deliver presents to children in every town. Thus, the probability that Santa *can* deliver presents to children in every town is

$$1 - \frac{13}{32} = \frac{19}{32}.$$

Therefore, our answer is

$$32 - 19 = \boxed{13}.$$

ONE DOWN

Notice that this pattern is simply the Koch snowflake (Mister K is Koch himself!). Intuitive from the problem statement, let K_0 be the area of the initial equilateral triangle.

We will start small. Notice that each new triangle has side length $\frac{1}{3}$ of the previous segment, so its area is $\frac{1}{9}$ of the area of a triangle built on the previous segment.

- **Iteration 1:** There are 3 sides, so 3 triangles are added, each with area $\frac{1}{9}K_0$. The total added area is

$$\frac{3}{9}K_0 = \frac{1}{3}K_0.$$

- **Iteration 2:** There are 12 sides, so 12 triangles are added, each with area $\frac{1}{81}K_0$. The total added area is

$$\frac{12}{81}K_0 = \frac{4}{27}K_0.$$

- **Iteration 3:** There are 48 sides, so 48 triangles are added, each with area $\frac{1}{729}K_0$. The total added area is

$$\frac{48}{729}K_0 = \frac{16}{243}K_0.$$

Thus, we find

$$K_3 = \left(\frac{1}{3} + \frac{4}{27} + \frac{16}{243} \right) K_0 = \frac{376}{243}K_0.$$

Now, growing big: from the iteration pattern above, the areas added at each step form a geometric series with first term $\frac{1}{3}K_0$ and common ratio $\frac{4}{9}$. Therefore,

$$\begin{aligned} K_\infty &= K_0 \left(1 + \sum_{n=0}^{\infty} \frac{1}{3} \left(\frac{4}{9} \right)^n \right) \\ &= K_0 \left(1 + \frac{\frac{1}{3}}{1 - \frac{4}{9}} \right) \\ &= \frac{8}{5}K_0. \end{aligned}$$

Computing the ratio,

$$\frac{K_3}{K_\infty} = \frac{\frac{376}{243}K_0}{\frac{8}{5}K_0} = \frac{235}{243}.$$

Then

$$2q - 2p = 486 - 470 = \boxed{16}.$$

TWO DOWN

To find the maximum possible length, we use the Polygon Inequality. Similar to the Triangle Inequality, the length of any single side must be strictly less than the sum of the remaining sides. Since four sides are already known, we add them:

$$5 + 7 + 12 + 15 = 39,$$

which gives an upper bound. If the fifth side were exactly 39, the resulting polygon would collapse into a line and thus be degenerate. Therefore, the largest possible length of the fifth side is

$$\boxed{38},$$

the greatest integer strictly less than the upper bound.

THREE ACROSS

To determine the color of each new house, it is enough to know whether the previous two houses have the same color or different colors.

Let A_n be the number of valid colorings of n houses in which the last two houses have the same color, and let B_n be the number of valid colorings in which the last two houses have different colors. Define

$$T_n = A_n + B_n.$$

If the last two houses have the same color, then the next house must be a different color, so

$$A_n = B_{n-1}.$$

If the last two houses have different colors, then the next house can be colored in two ways, so

$$B_n = 2A_{n-1} + 2B_{n-1}.$$

Combining these relations gives the recurrence

$$T_n = 2T_{n-1} + 2T_{n-2}.$$

The initial values are

$$T_1 = 3 \quad \text{and} \quad T_2 = 9.$$

Therefore, the number of valid decorations for 6 houses is 492, and our answer extraction gives

$$4 \cdot 9 \cdot 2 - 4 = \boxed{68}.$$